

E-COMMERCE CONVERSION DYNAMICS: CUSTOMER BEHAVIOR & PURCHASE INTENT ANALYSIS

Customer Behavior, Conversion
Dynamics & Purchase Intent



TOOLS USED

Python, Pandas, Numpy, Matplotlib, Plotly &
ML libraries

PRESENTED BY

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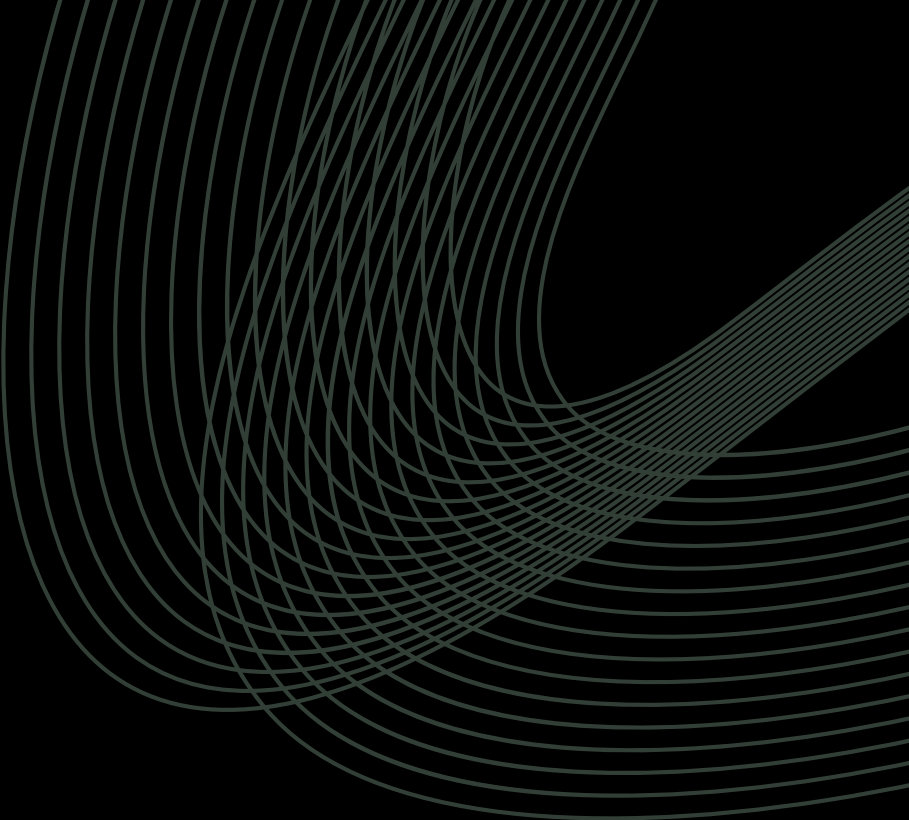
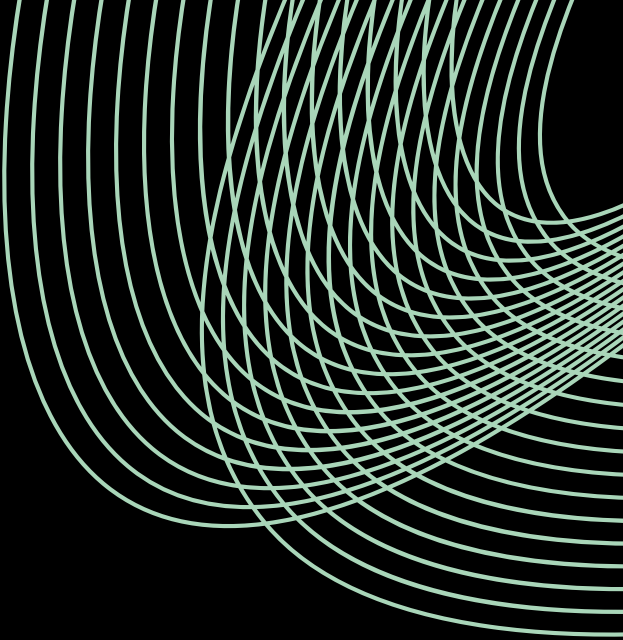


TABLE OF CONTENTS

Overview & Business Objective	3
Data Understanding & EDA	4
Data Preparation & Feature Engineering	5
Model Development & Experimentation	6
Model Evaluation & Comparison	12
Explainability & Final Model	34
Business Insights & Recommendations	36
Conclusion	37



Overview & Business Objective



1.1 PROJECT OVERVIEW

E-commerce platforms operate in a dynamic environment where purchase decisions are influenced by product engagement, traffic sources, visitor type & session behavior.

Understanding these behavioral patterns helps businesses identify sessions associated with stronger purchase conversion and better understand online customer activity.

The E-Commerce Conversion Dynamics project uses session-level online shopping data and machine learning to analyze purchase behavior and predict whether a session results in a completed purchase.

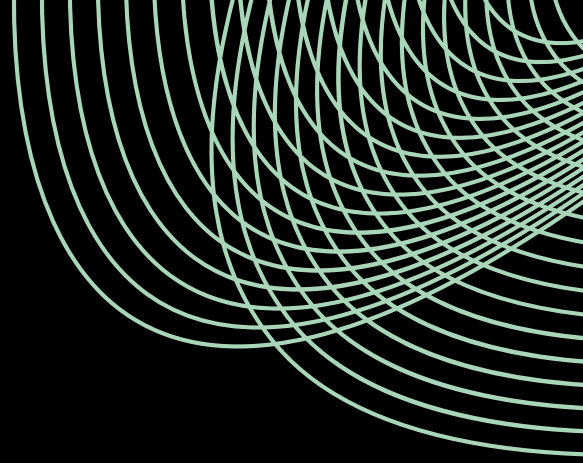
The project combines:

- Data understanding and cleaning
- Exploratory Data Analysis
- Behavioral feature engineering
- Class imbalance handling
- Machine learning classification
- Model evaluation and comparison
- Prediction-error analysis
- Model explainability
- Business-oriented insights

The objective is not only to predict purchase conversion accurately, but also to understand the behavioral signals behind the model's predictions and how these insights can support better e-commerce decision-making.



1.2 PROBLEM STATEMENT



Purchase conversion is influenced by multiple interacting aspects of an online shopping session, making it difficult to identify potential purchases using individual behavioral indicators alone.

The key problem addressed in this project is:

How can session-level e-commerce data be used to build a machine learning model that accurately identifies purchase sessions while also understanding the key behavioral factors associated with conversion?

The project therefore treats **Revenue** as the target variable and uses browsing behavior, product engagement, page value, traffic sources, visitor characteristics, and technical session information as predictive features.



1.3 PROJECT OBJECTIVES

The major objectives of the project are:

- Understanding the structure, quality, and characteristics of the online shopping dataset.
- Analyzing purchase conversion patterns across months, visitor types, traffic sources, and session behavior.
- Identifying relationships between browsing activity, product engagement, and purchase outcomes.
- Creating meaningful behavioral features and preparing the dataset for machine learning.
- Addressing class imbalance and developing multiple classification models for purchase prediction.
- Tuning and comparing classification models using appropriate performance metrics.
- Evaluating classification thresholds and investigating false-positive and false-negative predictions.
- Explaining model predictions using feature importance and permutation importance.
- Translating analytical and modeling results into practical e-commerce business insights.



1.4 PROJECT SCOPE

The project follows a structured workflow covering data analysis, behavioral feature engineering, machine learning, model evaluation, and explainability to understand purchase patterns, identify key conversion signals, evaluate predictive performance, and derive actionable e-commerce insights.

01

Data Analysis

- Dataset inspection
- Data quality assessment
- Purchase conversion patterns
- Behavioral relationship analysis

02

Conversion Modeling

- Feature preparation
- Behavioral feature engineering
- Classification modeling
- Model tuning and comparison

03

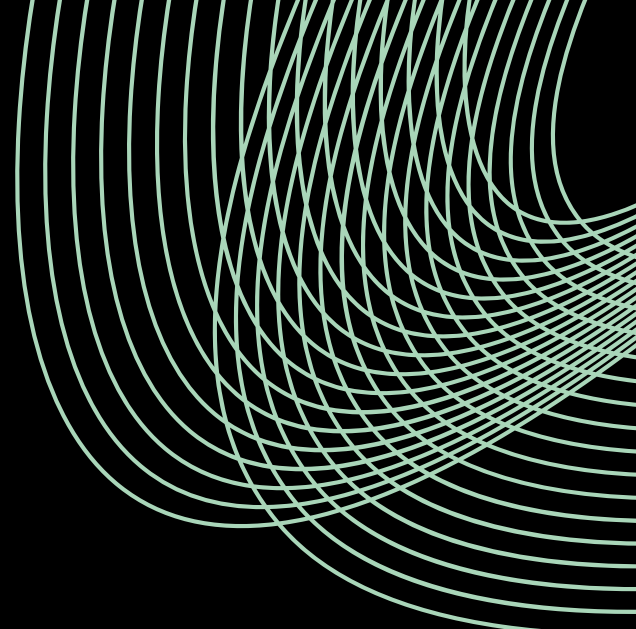
Explainability

- Feature importance analysis
- Original-feature aggregation
- Permutation importance analysis

04

Business Interpretation

- Purchase conversion observations
- Customer behavior insights
- Conversion signal interpretation
- Recommendations for e-commerce optimization



Data Understanding & Exploratory Data Analysis



2.1 DATA OVERVIEW

The dataset provides a comprehensive view of e-commerce shopping sessions, covering browsing activity, page engagement, visitor characteristics, technical attributes, traffic sources, special-day proximity, and purchase outcomes. These variables provide the foundation for analyzing customer behavior, identifying factors associated with purchase conversion & developing predictive models for e-commerce purchase intention.

Data Source: UCI

Dataset Size

12,330

Records

18

Original Features

Purchase Diversity

9

Regions

3

Visitor Type

20

Traffic Type



DATA OVERVIEW

Below is a detailed description of the feature set:

Dataset Features	Type	Feature Description
Administrative	Numerical (Discrete)	Number of administrative pages visited.
Administrative_Duration	Numerical (Continuous)	Time spent on administrative pages.
Informational	Numerical (Discrete)	Number of informational pages visited.
Informational_Duration	Numerical (Continuous)	Time spent on informational pages.
ProductRelated	Numerical (Discrete)	Number of product-related pages visited.
ProductRelated_Duration	Numerical (Continuous)	Time spent on product-related pages.
BounceRates	Numerical (Continuous)	Average bounce rate during the session.
ExitRates	Numerical (Continuous)	Average exit rate during the session.
PageValues	Numerical (Continuous)	Average value of pages visited.
SpecialDay	Numerical (Continuous)	Proximity to a special shopping day.
Month	Categorical	Month of the shopping session.
OperatingSystems	Categorical	Operating system used by the visitor.
Browser	Categorical	Browser used by the visitor.
Region	Categorical	Geographic region of the visitor.
TrafficType	TrafficType	Type of traffic source.
VisitorType	Categorical	Type of visitor.

DATA OVERVIEW

Below is a detailed description of the feature set:

Dataset Features	Type	Feature Description
Weekend	Categorical (Binary)	Indicates whether the session was on a weekend.
Revenue	Categorical (Binary)	Indicates whether a purchase was completed.
Product ID	Categorical	Unique identifier of the product
Category	Categorical	Product category
Region	Categorical	Geographic region of the store
Inventory Level	Numerical (Discrete)	Inventory available at the time of observation
Units Sold	Numerical (Discrete)	Number of units sold
Units Ordered	Numerical (Discrete)	Number of units ordered
Price	Numerical (Continuous)	Selling price of the product
Discount	Numerical (Discrete)	Discount applied to the product
Weather Condition	Categorical	Weather condition during the observation
Promotion	Binary	Indicates whether a promotion was active
Competitor Pricing	Numerical (Continuous)	Price of the comparable product offered by competitors
Seasonality	Categorical	Seasonal classification of the observation
Epidemic	Binary	Indicates whether an epidemic-related condition was present
Demand	Numerical (Discrete)	Target variable representing product demand

2.2 TARGET VARIABLE

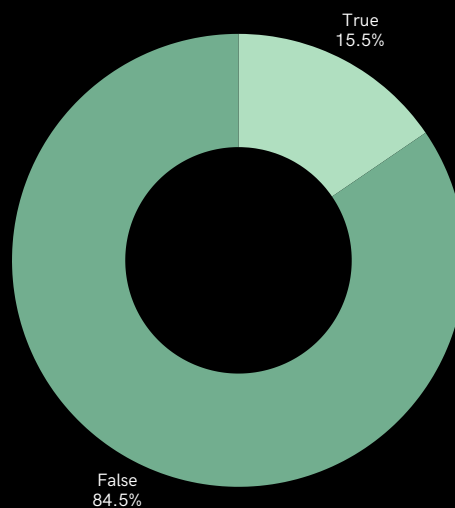
The target variable selected for prediction is:

Revenue

It represents whether the session resulted in a completed purchase.

True: *Purchase*

False: *No Purchase*



The target is imbalanced, with purchase sessions representing a smaller proportion of the overall dataset. This imbalance was therefore considered during model development.

